

Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer



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Quantum Computers for Space Exploration

Quantum Fisher information as local distinguishability

How does a small parameter change affect the state and its measurement statistics?

Nearby parameter values



Each measurement basis reveals a different statistical response.

Classical Fisher information

$$\mathcal{F}_C(\theta) = \sum_y \frac{[\partial_\theta p_y(\theta)]^2}{p_y(\theta)}$$

Quantum measurement limit

$$\mathcal{F}_Q(\theta) = \sup_{\{M_y\}} \mathcal{F}_C(\theta)$$

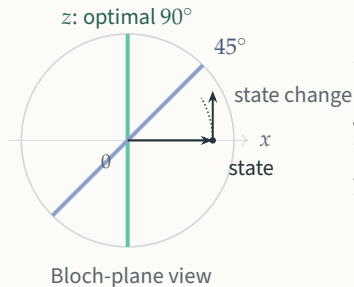
The same local geometry describes changes produced by trainable circuit parameters.

Thesis: Mathematical Background. Single-parameter, local estimation; schematic outcome distributions.

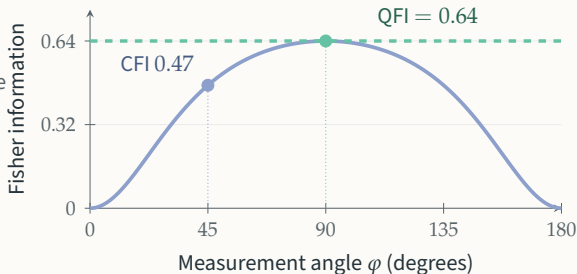
Measurement direction and accessible information

One mixed qubit and one unknown rotation angle; only the measurement basis changes.

Measurement axes



Information extracted



Changing the measurement raises CFI up to the quantum limit set by the state family.

Exact illustration: Bloch radius $r = 0.8$, local estimation of rotation θ (radians) at $\theta = 0$. Derivation: Appendix J.

Variational spectral estimation

Variational Quantum State Eigensolver (VQSE): a trainable spectral decomposition.

PCA: covariance matrix

$$C = \sum_i s_i \mathbf{v}_i \mathbf{v}_i^\top$$

Directions ranked by variance.

VQSE: density matrix

$$\rho = \sum_i \lambda_i |v_i\rangle \langle v_i|$$

Components ranked by probability weight.

$$V(\boldsymbol{\alpha}) \rho V^\dagger(\boldsymbol{\alpha}) \approx \text{diag}(\lambda_1, \lambda_2, \dots)$$

Frequencies estimate λ_i ; the inverse rotation $V^\dagger|i\rangle$ recovers eigenvectors.

Implemented VQSE supplies spectral information for fidelity-based QFI bounds.

Cerezo et al., npj Quantum Information 8, 113 (2022); Beckey et al., Phys. Rev. Research 4, 013083 (2022).

QFI geometry for variational optimization

Circuit controls α define an ansatz state family ρ_α .

For an objective $C(\alpha)$, account for how each control changes the state.

Newton's method

$$\Delta\alpha = -\eta H_C^{-1} \nabla C$$

$$H_C = \nabla^2 C$$

Curvature of the objective.

Quantum natural gradient

$$\Delta\alpha = -\eta (F^{\text{ans}} + \lambda I)^{-1} \nabla C$$

Ansatz QFIM F^{ans}

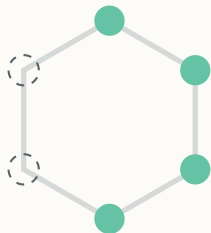
Distinguishability of neighbouring states.

QNG preconditions the gradient with state geometry rather than the objective Hessian.

Thesis: Mathematical Background; VQSE. $\lambda > 0$ regularizes QNG; Newton expression assumes an invertible Hessian.

Preparation and restricted access in the Ising model

Six interacting spins encode the transverse field; measurements access four of them.



$N = 6$ $n = 4$ accessible spins
Periodic boundaries

$$H(h_x) = -J \sum_j Z_j Z_{j+1} - h_x \sum_j X_j$$

Two preparation families

$$\rho_G = |\psi_0(h_x)\rangle\langle\psi_0(h_x)|, \quad \rho_\beta = \frac{e^{-\beta H(h_x)}}{Z_\beta}$$

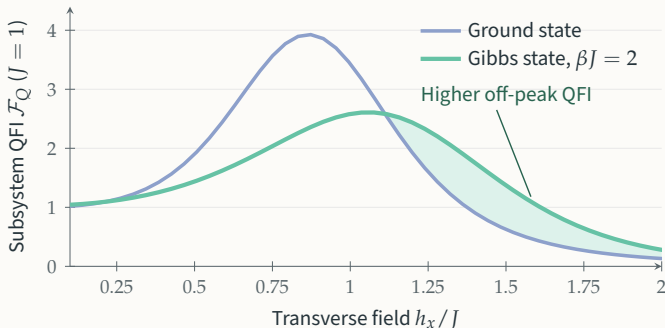
In both cases, the accessible state is
 $\rho_S(h_x) = \text{Tr}_E \rho(h_x).$

Compare the information in the same subsystem, across the same field range.

Thesis: Noise Analysis, periodic $N = 6$, $n = 4$ preparation comparison. Gibbs preparation precedes partial trace.

Thermal sensitivity beyond the ground-state optimum

A lower peak can coexist with greater sensitivity away from the optimum.



At $\beta J = 2$

Peak QFI:

thermal 2.61

ground state 3.93

Higher sensitivity
away from the
ground-state peak.

The preferred preparation depends on the expected field range as well as peak sensitivity.

Exact SLD calculation; periodic $N = 6$, $n = 4$, $J = 1$. Different preparation families; no universal thermal advantage.

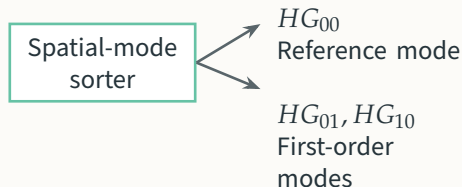
SPADE detector



Spatial-mode measurements and the available record

Spatial-mode sorting measures the displacement response of a faint companion.

Physical measurement



A choice of optical basis, before the data are recorded.

Archived observations

$$n_1 = n_{HG_{01}} + n_{HG_{10}}$$

525 settings $\times$ 100 repetitions

52,500 counts; 10 ms per repetition

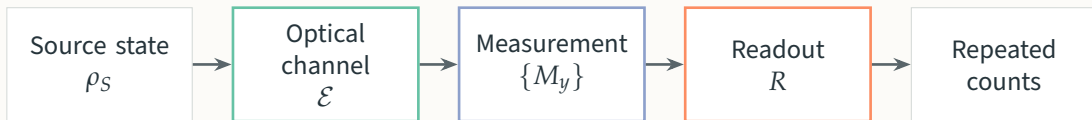
Missing: resolved and failure outputs;
incident-flux monitoring;
measurements in additional bases.

The record constrains the classical response; it does not identify the full optical state.

Thesis: Quantum Channel Model for SPADE. Experiment: Santamaria, Sgobba and Lupo, Optica Quantum 2, 46–56 (2024).

A quantum-to-classical channel model

A physical forward model for propagation, measurement and detector reporting.



$$p_y = \text{Tr}[M_y \mathcal{E}(\rho_S)] ,$$

$$p_{\text{rep}} = R p$$

Quantum stages

Loss, dephasing and coherent mixing.

Classical stages

Readout, calibration and count noise.

The framework preserves physical constraints and locates losses of information.

Thesis: sequential channel model. Coherence and optical-channel parameters are not reconstructed from aggregate counts.

Kraus representations are not unique

Different operator sets can describe exactly the same quantum channel.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad \sum_a K_a^\dagger K_a = I$$

Example: complete dephasing of a two-mode state in the chosen basis.

Basis projectors

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

Random phase flips

$$\tilde{K}_0 = \frac{I}{\sqrt{2}}, \quad \tilde{K}_1 = \frac{Z}{\sqrt{2}}$$

$$\sum_a K_a \rho K_a^\dagger = \frac{\rho + Z\rho Z}{2} = \text{diag}(\rho)$$

Kraus freedom changes the representation. Physical hypotheses concern the channel.

Two-mode illustration of the thesis HG-dephasing model at $v = 0$. Equivalent representations cannot be distinguished by system-only data.

A physical construction: mode-dependent loss

Mode transmission $0 \leq \tau_j \leq 1$; loss is represented by an orthogonal failure state.

Choose operators with explicit physical roles

$$K_t = \sum_j \sqrt{\tau_j} |j\rangle \langle j|, \quad K_j^f = \sqrt{1 - \tau_j} |\text{fail}\rangle \langle j|$$

Transmission and failure jointly account for every input photon.

$$K_t^\dagger K_t + \sum_j (K_j^f)^\dagger K_j^f = \sum_j [\tau_j + (1 - \tau_j)] |j\rangle \langle j| = I$$

Kraus form ensures complete positivity; completeness proves trace preservation.

Channel validity can be proved mathematically; physical suitability requires calibration.

Derived from the thesis mode-dependent loss map. Input: one-photon mode space after common survival; failure is retained in the output.

Information retained by the optical receiver

Compare quantum and classical information at successive stages of the receiver.

$$\mathcal{F}_Q(\rho_S) \geq \mathcal{F}_Q[\mathcal{E}(\rho_S)] \geq \mathcal{F}_C(\mathbf{p}) \geq \mathcal{F}_C(R\mathbf{p})$$

For channels and measurements independent of the parameter being estimated.

Checkpoint	Fraction of source QFI
● Resolved reference, first-order and failure outputs	99.68–100%
● Binary readout; assumed 0.35% port confusion	0.13–57.0%

Aggregation and readout confusion can strongly reduce the accessible information.

Conditional model: same nine settings, per photon surviving common loss. Ranges across settings; not reconstructed experimental QFI.

From state information to experimental access

Sensitivity depends on preparation, accessible degrees of freedom and measurement.

- Variational estimation

Implemented spectral methods for mixed-state QFI bounds.

QFI geometry also provides a direction for circuit optimization.

- Preparation over a field range

Thermal preparation offers higher off-peak sensitivity in the studied regime.

- A physically constrained observation model

Physical channels connect information limits to data and guide calibration.

Next: resolved SPADE outputs, independent calibration and a programmable modal basis.

QNG is a methodological connection; the optical variational protocol VQ-SPADE remains a proposal for a future experiment.

Spectral truncation and the local-QFI limit

Dominant eigencomponents bound a fidelity quantity at finite parameter displacement.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

F is root fidelity.

The retained eigenspace is state dependent; it is not a fixed detector-mode cut-off.

At regular operating points, $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$ as $\Delta \rightarrow 0$.

Finite-step bounds need not be pointwise bounds on the exact SLD QFI.

Thesis: Methods; Mathematical Background. Cerezo et al. (2022); Beckey et al. (2022).

The ansatz metric and its scope

The physical sensing parameter and the trainable circuit controls define different derivatives.

$$F_{\mu\nu}^{\text{ans}} = 4 \operatorname{Re} [\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]$$

Pure ansatz: $F^{\text{ans}} = 4g_{\text{FS}}$, the Fubini–Study metric up to convention.

$$\min_{\Delta\alpha} \nabla C^\top \Delta\alpha + \frac{1}{2\eta} \Delta\alpha^\top (F^{\text{ans}} + \lambda I) \Delta\alpha$$

Stationarity gives the regularized natural-gradient update.

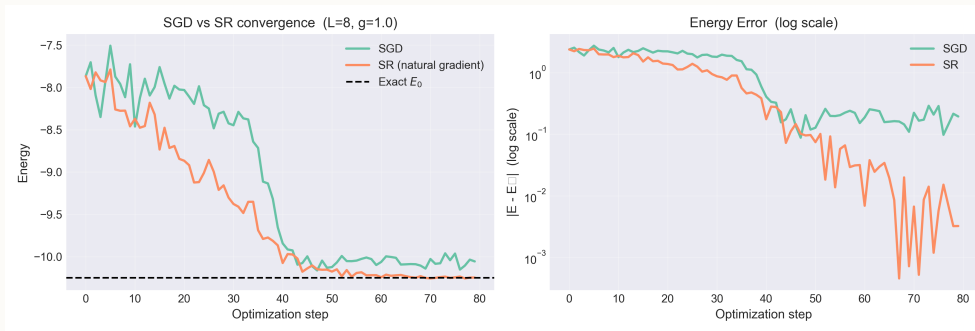
The quadratic term constrains state change, not objective curvature.

Mixed VQSE inputs require a mixed-state metric; a pure-state metric is otherwise a surrogate.

Thesis: Mathematical Background; VQSE. A full metric has quadratic parameter-count overhead; no general training speedup is claimed.

Stochastic reconfiguration in a neural quantum state

A separate pure-state RBM example illustrates metric-based optimization.



SR lowers the energy error in this run; it does not benchmark mixed-state VQSE.

Thesis figure: VQSE chapter. TFIM at $h_x/J = 1$; 80 steps, 300 Monte Carlo samples/step, $\eta = 0.05$, SR $\lambda = 10^{-4}$; same seed.

Thermal sensitivity and data processing

The Gibbs and ground-state families need not have the same field response.

$$\rho_{\beta}(h_x) = \frac{e^{-\beta H(h_x)}}{\text{Tr } e^{-\beta H(h_x)}}$$

The field changes both eigenvectors and thermal populations.

Partial trace still obeys data processing within each family:

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h_x)] \leq \mathcal{F}_Q[\rho_{\beta}(h_x)]$$

For an uncertain field, choose an operating range or prior.

These curves do not establish a universal Bayesian or minimax optimum.

The comparison changes the preparation family; it does not reverse a fixed noise channel.

Proof of Kraus-unitary freedom

A change of Kraus basis leaves the channel unchanged.

$$\tilde{K}_a = \sum_b u_{ab} K_b, \quad U^\dagger U = I$$

$$\sum_a u_{ab} u_{ac}^* = \delta_{bc} \quad \text{Orthonormal columns of } U.$$

$$\sum_a \tilde{K}_a \rho \tilde{K}_a^\dagger = \sum_{b,c} \delta_{bc} K_b \rho K_c^\dagger = \sum_b K_b \rho K_b^\dagger$$

Isometry in the Kraus index leaves the action on every input unchanged.

A physical realization adds assumptions about the environment; the reduced map alone does not select them.

Standard Kraus equivalence: an isometry, or a unitary after padding with zero operators. Algebra shown explicitly for this presentation.

A classical response does not fix a quantum channel

Embedding a stochastic map is a model construction, not process tomography.

$$\mathbf{p} \mapsto \text{diag}(\mathbf{p}), \quad \mathbf{p}' = R\mathbf{p}$$

For a column-stochastic R , one valid quantum extension is

$$K_{ji} = \sqrt{R_{ji}}|j\rangle\langle i|, \quad \mathcal{E}_R(\rho) = \sum_{ji} R_{ji}\rho_{ii}|j\rangle\langle j|$$

This extension measures and prepares in the chosen basis, erasing coherences.

For $R = I$, both the identity channel and complete dephasing reproduce the same populations.

Different channels can agree on the measured populations; this is distinct from Kraus freedom of one channel.

Thesis: classical/quantum embedding and HG coherence discussion. Channel-extension example derived for the presentation.

The loss map and its physical checks

The input mode space is embedded in an output space containing an orthogonal failure flag.

$$\mathcal{L}_\tau(\rho) = K_t \rho K_t^\dagger + \text{Tr}[(I - K_t^\dagger K_t)\rho] |\text{fail}\rangle \langle \text{fail}|$$

For any ancillary system, Kraus form preserves positivity:

$$(\mathcal{L} \otimes \text{id})(X) = \sum_a (K_a \otimes I) X (K_a^\dagger \otimes I) \geq 0 \text{ for } X \geq 0.$$

Boundary checks: $\tau_j = 1$ transmits mode j ; $\tau_j = 0$ sends it to failure.

Normalization: transmission plus failure has unit trace.

Physical validation: mode-injection measurements determine the τ_j .

Renormalizing the transmitted branch alone changes the question to postselected performance.

Thesis: mode-dependent loss and flagged mode reduction. The record lacks independent modal-transmission calibration.

Predictive calibration of the observed response

A physical mean with a Gaussian-process correction predicts held-out setting means.

$$\mu(d_a, \epsilon) = \mu_0(d_a, \epsilon) \exp[f(d_a, \epsilon) + \zeta]$$

Grouped five-fold validation; all components refitted within each training fold.

Held-out structure	RMSE	Standardized SD	95% coverage
Checkerboard cells	0.606	1.004	93.52%
Distance rows	0.734	1.041	94.67%
Source-ratio columns	0.609	1.009	93.71%

The validated target is the mean of 100 repetitions; this does not identify a complete quantum channel.

Thesis: Exoplanet Experiment Appendix, hierarchical calibration validation. RMSE in counts per 10 ms.

From SPADE to a variational optical measurement

VQ-SPADE is a proposed optical realization of variational state eigensolving.

$$p_k(\boldsymbol{\alpha}; d_a) = \langle HG_k | V(\boldsymbol{\alpha}) \rho(d_a) V^\dagger(\boldsymbol{\alpha}) | HG_k \rangle$$

A programmable coherent mode transformation provides the trainable basis.

Resolved outputs, flux and failure events provide normalized probabilities.

Nearby states and additional overlap measurements support fidelity-based QFI bounds.

The present record motivates the acquisition design; it does not implement this protocol.

Thesis: proposed VQ-SPADE scheme. Additional independent channel and background calibrations are required.

An exact example of measurement-dependent CFI

Unknown parameter: state rotation θ . Measurement control: basis angle φ .

$$\rho_\theta = \frac{I + r(\cos \theta X + \sin \theta Z)}{2}, \quad r = 0.8$$

$$M_\pm^{(\varphi)} = \frac{I \pm (\cos \varphi X + \sin \varphi Z)}{2}, \quad p_\pm = \frac{1 \pm r \cos(\theta - \varphi)}{2}$$

$$\mathcal{F}_C(\varphi)|_{\theta=0} = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi} \leq r^2 = \mathcal{F}_Q$$

At $\theta = 0$, the SLD is $L_0 = rZ$, so measuring Z attains QFI.

The 45° measurement gives $\mathcal{F}_C = 8/17 \simeq 0.47$, compared with $\mathcal{F}_Q = 0.64$.

These directions label measurements for one parameter, not directions in a two-parameter QFIM.

Derived from the CFI and SLD definitions in the thesis Mathematical Background. $r < 1$ keeps this example full rank and regular.